

At this presentation, we began with solar energy and the limits of relying on it alone. he pointed out that the sun is not always shining, clouds reduce how much power we get, and storing energy at scale is still very expensive. He mentioned pumped storage as one option, where water is moved uphill when there is extra electricity and then run back down through turbines later. In his view, the biggest cost of an economy built on clean energy is not the energy itself but the enormous investment needed to build a grid that can move power around in a smart way. He also spoke about how climate change is largely political in the United States and stressed the need to reach people outside the usual “blue state” audiences with clear education and respectful conversations.

He then shifted into a personal story. He came from a modest high school, went to Amherst, and first majored in theatre. He played tennis but switched to squash when he realized everyone else was better. A strong chemistry teacher led him toward science, and he eventually tried physics. After about a year and a half he took an astronomy course and learned that, unlike many stars, the sun does not have a companion star. He found this fascinating. The course required an eight page paper, but he became so absorbed that he turned in twenty one pages. He took another astronomy class at Smith College, and this sealed his decision to become an astronomer, because he felt astronomy is one of the things that makes us distinctly human.

Later he spoke about teaching at Columbia and his love for the core curriculum, which focuses on the humanities. To balance that, Columbia created Frontiers of Science, and he noted that this is now its twenty second year. In describing astronomy courses, he talked about two tracks. Courses without an asterisk are meant for majors and use calculus. Courses with an asterisk are designed for students with no physics background and avoid heavy math. He argued against separating “real science people” from everyone else and said he dislikes the idea of one course for experts and another for non experts. Near the end, he mentioned serving for ten years as president of a small liberal arts university in Canada. There, every student followed a shared curriculum for the first two years and studied in an unusual one class per month system, which he clearly admired as a way to help students focus deeply on each subject.